MIL-C-18843E
4 June 1986
SUPERSEDING
MIL-C-18843D
11 April 1974

MILITARY SPECIFICATION

CUTTING MACHINE, OXYGEN-FUEL, MOTOR DRIVEN, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable, electric motor driven, oxygen-fuel cutting machine for cutting straight lines, arcs, and circles from metal plate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3433
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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - *Preferred Metric Units for General Use by the Federal Government.*

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/RMA/CGA IP-7 - Specification for Rubber Welding Hose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

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COMPRESSED GAS ASSOCIATION (CGA)

CGA Pamphlet E-1 - Standard Connections for Regulator Outlets, Torches and Fitted Hose for Welding and Cutting Equipment.

(Application for copies should be addressed to Compressed Gas Association, Inc., 500 Fifth Avenue, New York, NY 10110.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of cutting material in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, pressure gauges, and other similar devices. When only one system of graduation is acceptable, the particular graduation

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required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the cutting machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. All safety devices normally incorporated in the machine by the manufacturer for the protection of the operator shall be furnished. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.5).

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation (see 6.5).

3.2.8 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible and identifiable.

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

3.4 Components.

3.4.1 Carriage assembly. The carriage assembly shall consist of a housing, drive motor, drive train, controls, and other features necessary to meet the requirements of this specification. The carriage housing shall completely enclose the drive components. The carriage shall be mounted on two drive and two caster wheels. The caster wheels shall be free to pivot through an angle of not less than 180 degrees about a vertical axis and shall have a lock for straight line cutting. A torch mounting arm with provision for adjusting the torch in the horizontal and vertical direction shall be attached to the top of the carriage housing. The housing shall provide for mounting of the radius rod assembly.

3.4.1.1 Drive motor. The drive motor shall be of sufficient size and horsepower to meet maximum operating conditions.

3.4.1.2 Drive train. The drive train shall consist of driving wheels connected to the drive motor through a clutch. The clutch shall disengage the drive motor from the driving wheels. The driving wheels shall be of steel and shall have a knurled periphery.

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3.4.1.3 Controls. Controls shall be furnished to change direction of carriage travel, speed of carriage travel, and to stop carriage travel. The controls shall be mounted in the carriage housing in a position convenient to the operator. Speed adjustment shall be possible while the carriage is in motion.

3.4.2 Torch mounting mechanism. The torch mounting mechanism shall provide for locating and locking the torch in the horizontal and vertical positions and in any angular position between the horizontal and vertical positions. The torch mounting mechanism shall consist of a torch holder, arm, mounting bracket, mounting post, and necessary clamps. The arm that attaches to the carriage and supports the torch shall be of sufficient strength to provide the accuracy specified in 3.5 when the torch is in any position. Lateral adjustment of the torch shall be not less than 8-3/4 inches. The torch shall have sufficient vertical adjustment to accommodate the cutting capacity specified in 3.5. The horizontal and vertical adjustments shall be accomplished by means of rack and pinion gears.

3.4.3 Cutting torch. The machine shall be furnished with an oxygen-fuel gas cutting torch. The torch shall be of the machine type with a cylindrical barrel, a rack for vertical adjustment, gas valves, and a cutting tip. The torch shall be of the two or three hose style designed for use with oxygen and acetylene, natural gas, or propane fuel gases. The end of the torch housing and the fuel and oxygen valves shall be manufactured of bronze or brass. The torch head shall be constructed to receive cutting tips of various sizes.

3.4.3.1 Cutting tips. Cutting tips shall be fabricated of copper or copper-tellurium and shall have a heat conductivity not less than 90 percent of pure electrolytic copper. Cutting tips shall be one piece, and unless otherwise specified designed for use with acetylene. The tips shall have not less than 4 preheat holes and one cutting oxygen hole. The tips shall be permanently and legibly identified for size according to the manufacturer's standard commercial practice. Cutting tip seats shall provide a tight seal to prevent gas leaks around the tip nut. The center cutting oxygen orifice shall provide a stream of oxygen which travels down the centerline of the preheat flame group and shall not deviate from the center line by more than 2 degrees. A set of tips capable of cutting low carbon steel from 1/16 inch to 6 inches thick shall be furnished. Additional tips shall be as specified (see 6.2.1).

3.4.4 Track. A steel track of the interlocking type, at least 60 inches long, shall be provided with each machine for straight line cutting. The track shall be of welded construction or shall be fabricated from a single piece of stock. A precision machined guide groove shall be provided in at least one rail of the track to provide guidance of the machine's movements and to assure conformance with the accuracy specified in 3.5. Additional lengths of interlocking track shall be furnished as specified (see 6.2.1).

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3.4.5 Radius rod assembly. A radius rod assembly shall be furnished with the machine for circle and arc cutting within the capacities specified.

3.4.6 Hoses and fittings. The machine shall be furnished with all hoses and fittings required for operation. All hoses furnished shall comply with ANSI/RMA/CGA IP-7. The fittings attached to the hoses shall conform to CGA Pamphlet E-1, connector numbers 020, 021, 022, or 023, as applicable. The hose fittings shall provide sufficient sealing to prevent gas leakage at all operating pressures. The length of the hose connecting the oxygen and fuel gas source to the machine shall be not less than 25 feet. When the machine is furnished with a hose manifold, 30 inch length of oxygen and fuel gas hose shall be furnished.

3.4.7 Electric power cord. A three conductor electric power cord not less than 25 feet in length shall be furnished with the machine. The power cord may be permanently attached to the cutting machine or may be connected by a receptacle and plug.

3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Unless otherwise specified (see 6.2.1), the machine shall be designed to operate from a 120 volt, 60 Hz, single phase power source.

3.5 Performance. The machine shall be capable of cutting SAE 1020 or other low carbon steel up to 6 inches thick. The machine shall be capable of cutting a straight line of not less than 60 inches with a deviation of not greater than ± 0.03 inches, circles ranging from not greater than 4 inches to not less than 54 inches in diameter. The machine shall have a cutting speed range from not greater than 4 inches per minute to not less than 32 inches per minute.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.7 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.8 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.9 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be

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applied directly onto the item by environmentally protected decalcomania transfer, a metal wrap-around tag, a stencil, silk screening, or by any other method suitable for the use intended. Markings shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and its accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The cutting machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be connected to the electric power source and to the oxygen and fuel gas supply. All components of the gas system shall be checked for leaks. Gas pressures shall be set in accordance with the supplier's recommendations. The torch shall be lighted and adjusted to obtain a neutral flame. The machine shall be operated to insure conformance with all requirements specified herein without performing any cutting operations.

4.7.2 Performance tests.

4.7.2.1 Depth of cut test. A sample of SAE 1020 or other low carbon steel 6 inches thick and 12 inches wide shall be cut. The time required for the 12 inch cut shall be not greater than two minutes. The cut edge of the

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kerf shall be square and the slag on the bottom of the cut shall be easily removable.

4.7.2.2 Speed of cut test. A sample of SAE 1020 or other low carbon steel 1/4 inch thick and 48 inches long shall be cut. The time for the cut shall be not greater than two minutes. Accuracy of the cut shall be as specified in 3.5. The cut edge of the kerf shall be square and the slag on the bottom of the cut shall be easily removable.

4.7.2.3 Circle cutting test. A circle cutting test shall be performed using a sample of SAE 1020 or other low carbon steel to demonstrate the machine's capability to cut minimum and maximum diameter circles specified in 3.5.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The gas cutting machine described in this specification is intended to be used to cut straight lines, circles, and arcs from metal plate.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, if required (see 3.1).

c. If the cutting machine design and construction is required to be in a specific measurement system (US or SI), state required system (see 3.2.1).

d. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

e. Additional safety and health requirements, if required (see 3.2.4).

f. Painting, if different (see 3.3.5).

g. Additional tips, if required (see 3.4.3.1).

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h. Additional lengths of track, if required (see 3.4.4).

i. Electrical system and input power, if different (see 3.4.8).

j. Additional performance features, if required (see 3.5.1).

k. Standard equipment, if different (see 3.6).

l. Specify and fully describe optional equipment, if required (see 3.7).

m. Repair parts, if required (see 3.8).

n. Nameplate, if different (see 3.10).

o. First article inspection, if different (see 4.4).

p. Quality conformance inspection, if different (see 4.5).

q. If preservation, packaging, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, wiring diagrams, and acceptance test reports, should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Cutting machines
Cutting tools
Flame cutting
Gas cutting
Metal cutting
Oxyacetylene cutting
Oxygen cutting
Torches

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6.5 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirements that will integrate the new machine into its future operating environment. The cutting machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial cutting machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (e).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Review activities:
DLA - GS
Air Force - 84

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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MIL-C-18843E	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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